Statement of Career Goals, Research Accomplishments, Teaching and Mentoring, and Proposed Research

Wentao Zhang · Research Assistant, Power Electronics and Machines Centre (PEMC), University of Nottingham

Provost's Postdoctoral Fellowship in Engineering, University of Notre Dame · Proposed primary mentor: Prof. Nishanth Gadiyar

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Career goals

My goal is a tenure-track faculty post in **electrical machines and physics-informed AI**, leading a group that designs, builds and tests machines and treats learned models as instruments that must agree with measurement. Electrification of transport and industry depends on machines that are smaller, more efficient and less dependent on critical materials. Designing them is a coupled electromagnetic, thermal and mechanical problem, and the engineers who will use AI for it need models whose errors are known. I want my group to provide such models and to train engineers who can judge them.

In the longer term I see three lines of work for such a group: learned models that are consistent with the physics of the machine, models that transfer to new topologies and materials so that designers are not bound to one machine family, and digital twins of machines and drives that are updated from measured data. The fellowship concentrates on the first two and lays the measured foundation for the third, so that I would begin a faculty post with a defined programme, preliminary results and collaborators already in place.

My PhD and industrial work gave me the full chain from specification to dynamometer test, and since September 2025 I have written an in-house finite element (FE) code at PEMC and built the first physics-informed field models on top of it. The gaps in my preparation are equally clear. I have not yet led a funded research project or formally supervised a student. My funding record is at an early stage, consisting of an N-ZEEE Flexible Funding award of £2,700 (2026) to present [C2] at IEEE ICEMS 2026. My research network is in China and the United Kingdom rather than the United States.

The fellowship addresses these gaps directly. It would give me an **independent research programme** of my own design for 24 months; mentoring from Prof. Gadiyar, whose route through WEMPEC, GE Research and Oak Ridge National Laboratory connects university, industrial and national-laboratory research in the US; the opportunity to mentor graduate and undergraduate students formally; practice in preparing proposals to NSF, DOE and ARPA-E alongside my mentor; and the College of Engineering Postdoctoral Training program, with individual professional development from the Director of Postdoctoral Studies and a cohort of fellows. By the end of the fellowship I aim to have a coherent body of journal papers on physics-consistent learned models, a validated test case shared with the AMPERE Lab, at least one proposal developed with my mentor, and a mentoring record on which to base applications for faculty posts. Notre Dame offers the combination I need for this step: a new machines laboratory with hardware and a stated AI-for-design thrust, a College with a structured postdoctoral training programme, and a university strategy that places data, AI and sustainability at its centre.

Previous research accomplishments

I completed my PhD in Electrical Engineering at Southeast University (SEU), Nanjing, in September 2025 under Prof. Zhongze Wu, with a dissertation on the multi-physics analysis and design of hairpin-winding permanent magnet (PM) traction machines. Since then I have been a Research Assistant at PEMC, University of Nottingham, within the EPSRC-funded AIMD programme (Nottingham, Bristol, Newcastle), working on magnet-free synchronous machines. My record comprises **26 journal papers (20 in IEEE Transactions, 11 as first author)**, 19 conference papers and 9 granted patents. My first-author paper on *n*-layer flat-wire winding design [J5] received the **2025 IEEE Transactions on Transportation Electrification Prize Paper Award (2nd Prize)**, and I also received 1st Prize in the 2025 IEEE TEC Electric Vehicle Design Competition and the 2025 MDPI *Machines* Best PhD Thesis Award. Four strands of this work lead directly to the proposal.

Multiphysics design taken to hardware. In an industrial programme with Inovance, a world-leading EV traction machine supplier, I was technical lead for two high-speed oil-cooled PM traction machines, taken through specification, multi-objective design, drawing and tolerance release, supplier manufacture and AVL

dynamometer test. The measured results were 270 kW, 380 Nm, 7.87 kW/kg and 97.4% peak efficiency at 20,000 rpm, and 113 kW, 138 Nm, 7.78 kW/kg and 95.3% peak efficiency at 18,000 rpm. The programme left me with efficiency maps, thermocouple records and current waveforms from built machines, the kind of measured ground truth against which the proposed models will be checked.

Models validated against measurement. I developed an oil-cooled lumped-parameter thermal network **validated against thermocouples across the full operating map** [J10], building on a transient thermal model [J12] and a patented thermal-network method [P4]. Rotor stress relieving grooves for high-speed operation were confirmed by a 22,500 rpm overspeed test [J11], and I compared bridge-retained and carbon-fibre-sleeved interior PM rotors in [C2]. With colleagues I quantified the effect of PWM excitation on loss and temperature [J22] and of glued versus welded stator cores on vibration [J23], and I analysed 3D end effects in ANSYS Maxwell and JMAG [J9]. The *n*-layer flat-wire winding methodology [J5] was built around hairpin forming limits. These studies showed me where simple models break down and how far a prediction can sit from measurement.

Fast design with AI and physics. A hybrid AI-plus-physics optimisation [J6] cut a 67-hour FE optimisation to 2.3 hours, and a patented dual-model database method [P3] pairs a fast model with high-fidelity FE. At PEMC I wrote an **in-house Python FE package** (geometry, mesh, electromagnetic, structural-modal and stress solvers) and trained on its data a physics-informed full-field magnetic surrogate of an interior PM machine [C1]. Instead of a few scalar outputs it predicts the full magnetic vector potential from 15 design inputs, so torque and iron loss follow from standard FE post-processing. On unseen designs it reaches $R^2 = 0.9996$ for torque and 0.9730 for iron loss, more than 94% faster than FEA. I also lead a six-month project on LLM-agent-assisted design and training of physics-informed surrogates, supported by in-kind API credits from the OpenAI Researcher Access Program.

Machines with fewer or no magnets. Much of my PhD and master's research concerned wound-field and hybrid-excitation switched flux machines, which use no rare-earth magnets or less of them. This work covered their DC-winding induced voltage and current dynamics [J1], [J4] and the trade-off between flux regulation and induced voltage [J7], and my current AIMD work is on magnet-free synchronous machines. My axial-flux experience is limited to co-authored magnetic-equivalent-circuit modelling [J21], [J25] and one patent [P2], and I have not worked on additively manufactured machines. I present both as design spaces I would extend into with the AMPERE Lab, not as established expertise. Taken together, these strands give me the three ingredients the proposal needs: an FE solver I control, measured data from built machines, and first evidence that physics-based training terms change what a learned model gets right.

I referee for *IEEE Trans. Industry Applications*, *IEEE Trans. Energy Conversion* and *IEEE Trans. Transportation Electrification*, and was Co-Organiser of the ICEM 2026 Special Session "Next-Generation Electrified Transport: Advances in Electric Drive Systems" (Funchal, Madeira, Portugal, September 2026) with Dr Lingyun Shao (NUAA).

Teaching and mentoring experience

From 2021 to 2025 I was **teaching assistant** for the SEU postgraduate course *Integrated Practice of Electrical Machine Systems*. I prepared and delivered tutorials, held Q&A and consultation hours, gave structured written feedback on students' final-project proposals and made mark recommendations to the course coordinator. I developed tutorial and assessment materials and two research-derived resources, a coupled electromagnetic-thermal modelling exercise and an embedded motor-drive practical. The lesson I took from that course is that students understand a model best when they have to check it against something real, and the same principle runs through the research proposed below.

My approach to teaching rests on three habits drawn from research practice. Every analytical or simulated result is compared with data, so that students learn judgement as well as procedure. Students should know what a finite element solver or a neural network is doing before they trust its output, which matters more now that AI tools produce plausible engineering answers on demand. Feedback should say what to do next, so that each assessment improves the next piece of work. As a faculty member I would expect to teach electrical machines and drives, power electronics and numerical methods, and to introduce machine learning for engineering systems at senior undergraduate or graduate level.

Within the SEU group I gave day-to-day technical guidance to junior PhD and M.Eng. researchers in finite element analysis, thermal-network construction and PWM-loss modelling, and several of those projects became

jointly authored papers in my publication list. I continue to mentor junior researchers informally at PEMC. I have not yet formally supervised a student, and my teaching portfolio is at an early stage.

During the fellowship I would build both deliberately. With Prof. Gadiyar's agreement I would **co-mentor graduate and undergraduate students** in the AMPERE Lab on well-defined parts of the project, such as an FE data-generation pipeline or a test-bench comparison, with written goals and regular review. I would contribute shared coding and validation practices so that every student's model is checked against FE and measurement in the same way. Where the department sees a need, I would be glad to give guest lectures in machines, drives or numerical methods courses, and I would use the Postdoctoral Training program to gain structured training in mentoring and teaching before applying for faculty posts.

Proposed research during the fellowship

Physics-consistent learned models for multiphysics design of electric machines beyond rare-earth magnets

Motivation. Designing an electric machine means solving coupled electromagnetic, thermal and mechanical problems, and each FE evaluation takes minutes to hours, so only a small part of the design space is explored. Learned surrogates are far faster, but two weaknesses limit their use in design. First, an accurate-looking field can hide inaccurate physics. In my current work (preliminary, unpublished, single training seed), a coordinate-network / neural-operator model predicts the magnetic vector potential at any point, conditioned on the design and rotor position. In that work **a 5% nodal field error hid a 70% energy-norm error**. Adding an FE energy-norm term to training, with differentiable B-H curves, reduced the energy-norm error from 29.8% to 5.4%, and a rotor-frame coherence term reduced the rotor iron-loss error from 50% to 13%. Second, most surrogates are tied to one topology and one set of materials, and changing either means regenerating the FE data. In the same preliminary work I transferred a trained model to a new magnet grade (remanence 1.165 to 1.42 T) by fine-tuning a frozen checkpoint, which suggests that the cost of moving to new materials and topologies can be reduced. The fellowship asks one question: *can a learned model of an electric machine be held to the physics it replaces, transferred to topologies and materials it was not trained on, and made accurate enough in loss and temperature to be trusted in design?* Owning the FE solver matters here, because I can compute energy and loss terms inside the training loop, which commercial tools do not expose, while JMAG provides an independent check.

Approach. The work rests on three choices. The first is to measure error where the designer feels it, in energy, flux linkage, torque, loss and temperature, computed with the same FE operators that produced the training data, since my preliminary results show that field accuracy alone is not a sufficient test. The second is to treat transfer as a measurable quantity, reporting how much new FE data a new topology or material needs, so that the cost of extending a model is known before it is attempted. The third is to close every loop against measurement, using machines that have been built and tested. Together these choices aim to turn a surrogate from a fast interpolator into a design instrument with a known error.

Aim 1 (months 1 to 8): a physics-consistent field operator. I will extend the preliminary model into a field operator trained with energy-norm and loss-consistent terms computed by my FE solver, so that the training objective measures error in the quantities a designer uses (stored energy, flux linkage, torque and iron loss) and not only in nodal field values. The operator will take design parameters, rotor position and spatial coordinates as inputs and return the vector potential, from which flux density, energy and loss follow through the FE post-processing chain. Because the output is defined at any point, the same model can be evaluated on meshes of different resolution, so its error can be compared directly with the solver's discretisation error. Held-out designs will be drawn from regions of the design space excluded from training, not only from random splits, so that extrapolation is tested explicitly. I will add calibrated uncertainty estimates and use them for active learning, so that the FE solver is called only where the model is unsure, and I will check the calibration on held-out designs. Key results will be repeated over several training seeds, since the current results come from one. Training data will cover at least two topology families, the interior PM machines of [C1] and a magnet-free wound-field family from my AIMD and PhD work. *Milestone (month 8):* energy-norm and iron-loss errors on held-out designs within a stated error band, set in month 2 against the FE solver's own mesh-refinement error, for at least two topology families; first journal paper submitted.

Aim 2 (months 6 to 16): transfer to new topologies and materials. Starting from a frozen checkpoint, I will fine-tune with small FE datasets in three directions: rare-earth-lean and lower-grade magnets, extending

the magnet-grade transfer above; **magnet-free and hybrid-excitation machines**, drawing on the AIMD programme and my wound-field background; and AMPERE Lab designs, namely multi-material additively manufactured SynRel rotors and axial-flux machines. Axial-flux machines need 3D or multi-slice models, so this last case tests whether the method extends beyond 2D cross-sections, and I would agree its scope with Prof. Gadiyar. The study will also establish which parts of the model transfer and which must be retrained. Each transfer will be compared with two baselines, a model trained from scratch on the new data and the unchanged source model, so that the benefit of transfer is measured rather than assumed. Design parameters that are new to the target, such as magnet remanence in the preliminary study, will be added as inputs and learned during fine-tuning. *Milestone (month 16):* for each case, the fine-tuned model reaches the accuracy of a model trained from scratch, and I report the fraction of the from-scratch FE data that transfer required; second journal paper submitted.

Aim 3 (months 12 to 24): coupled loss-thermal prediction and validation against measurement. Loss maps predicted by the operator will feed the validated thermal-network approach [J10], [J12], including PWM effects on loss [J22], giving temperatures across the operating map with uncertainty carried through from Aim 1. Predictions will be compared with my existing dynamometer data (efficiency maps and thermocouple records). With the AMPERE Lab I will design **one joint test case** in which the model predicts before the machine is measured. Thermal prediction is where small loss errors become decisive, because winding and magnet temperatures set the continuous rating and the demagnetisation margin, so the comparison will report temperature errors at the measured thermocouple locations as well as efficiency-map errors. Where model and measurement disagree, I will trace the disagreement to its source (field error, loss model or thermal parameter), which is itself useful information for designers. *Milestone (month 24):* validated multiphysics predictions on one AMPERE machine and one Inovance-derived dataset; third journal paper submitted.

Resources. The $15,000 research allowance would cover cloud or workstation GPU time and conference travel. The AMPERE Lab's machines and test facilities provide the hardware for Aims 2 and 3, and JMAG (a lab sponsor) provides an independent FE reference alongside my in-house solver. I bring the solver, the preliminary training framework and my dynamometer datasets, the last used within the terms of the original programme. If the 3D axial-flux case proves too costly within 24 months, Aim 2 will concentrate on the additively manufactured SynRel rotors and the axial-flux case will become the basis of a follow-on proposal. If industrial data cannot be used in a publication, the joint AMPERE test case will serve as the published validation.

Expected outcomes. The fellowship would produce three journal papers submitted to IEEE Transactions journals, conference papers at IEEE IEMDC or ECCE, a published evaluation protocol (energy-norm, loss and temperature metrics with held-out design splits) that other groups can apply to their own surrogates, and a validated test case shared with the AMPERE Lab. A model that transfers cheaply to rare-earth-lean and magnet-free machines lowers the cost of exploring designs that reduce dependence on critical materials, and the students who work on the project would learn to treat a learned model as something to be tested against physics and measurement.

Timeline.

- *Months 1 to 3.* Set up the solver and training pipeline at Notre Dame; agree metrics, error bands and AMPERE test cases with Prof. Gadiyar.
- *Months 1 to 8 (Aim 1).* Physics-consistent operator with calibrated uncertainty; milestone and first journal paper at month 8.
- *Months 6 to 16 (Aim 2).* Transfer studies across materials and topologies; milestone and second journal paper at month 16; conference presentation at IEEE IEMDC or ECCE.
- *Months 12 to 24 (Aim 3).* Loss-thermal coupling; joint test case designed in months 12 to 18 and measured in months 18 to 22; milestone and third journal paper at month 24.
- *Months 12 to 20.* Draft a proposal with Prof. Gadiyar to NSF, DOE or ARPA-E, building on Aims 2 and 3.
- *Throughout.* Co-mentoring of AMPERE Lab students and participation in the Postdoctoral Training program.

Primary mentor and rationale

I propose **Prof. Nishanth Gadiyar**, Assistant Professor of Electrical Engineering and leader of the AMPERE Lab, as my primary mentor. Prof. Gadiyar received his PhD from the University of Wisconsin-Madison (WEMPEC) in 2023, worked at GE Research from 2023 to 2024 and was a Weinberg Distinguished Staff Fellow at Oak Ridge National Laboratory from 2024 to 2026 before joining Notre Dame in 2026. He received

the 2026 IEEE IAS Andrew W. Smith Outstanding Young Member Award and is General Co-Chair of IEMDC 2029. I selected him for three reasons.

First, the research is closely aligned. The AMPERE Lab's **"AI-Accelerated Design"** thrust ("physics-informed machine learning and optimization to accelerate discovery") is the subject of this proposal, and its "Integrated Electro-Mechanical Systems" thrust ("co-design and optimization of magnetics, power electronics, thermal systems, and mechanical loads") is directly connected to the coupled loss-thermal work of Aim 3. His work on multi-material additively manufactured SynRel rotors, winding design for independent control of air-gap harmonics and additively manufactured axial-flux stators supplies the new topologies and materials on which a transferable model must be tested, and the lab's mission of machines "less dependent on critical materials" matches the aim of designing beyond rare-earth magnets.

Second, his background at WEMPEC, GE Research and Oak Ridge National Laboratory offers an **industrial and national-laboratory route** for the work, and would help me build the US research network I do not yet have. Third, the AMPERE Lab was launched in June 2026 and is still being built, so a fellow can help shape its AI thrust. In return I can bring an in-house FE solver, measured datasets from built machines and experience of taking machines from design to test; JMAG, which I use routinely, is also a lab sponsor. I would propose regular meetings, a written development plan agreed in the first month with Prof. Gadiyar and the Director of Postdoctoral Studies, and joint review of the milestones above. No co-mentor is named at this stage, and a co-mentor from Notre Dame's Data, AI, and Computing community could be added in discussion with Prof. Gadiyar.

Connection to Notre Dame's strategic framework

The proposal connects to three parts of Notre Dame's 2033 Strategic Framework. The **Data, AI, and Computing Initiative** "pursues an ambitious research and education agenda that balances ground-breaking foundational advances in data, AI and computing", and the question of when a learned model may stand in for a physical solver is a foundational one. The **Just Transformations to Sustainability Initiative** recognises that sustainability challenges cascade through "food, energy, water, and infrastructure", and machines that are more efficient and need fewer critical materials address the energy part of that chain. The work would also connect to Notre Dame Energy (ND Energy) and its aim of "a more sustainable energy future". I would be glad to discuss the proposal further and to contribute to the College's postdoctoral community.

Reference labels ([J], [C], [P]) follow the publication list in the attached CV.

References

[J1] **W. Zhang**, W. Hua, Z. Wu, G. Zhao, Y. Wang, and W. Xia, "Analysis of DC winding induced voltage in wound-field flux-switching machine with air-gap field modulation principle," *IEEE Trans. Ind. Electron.*, vol. 69, no. 3, pp. 2300–2311, Mar. 2022.

[J4] **W. Zhang**, Z. Wu, Y. Fan, W. Hua, and M. Cheng, "Analysis and improvement of DC winding current dynamics in wound field switched flux machine," *IEEE Trans. Ind. Appl.*, vol. 60, no. 4, pp. 6023–6032, Jul.–Aug. 2024.

[J5] **W. Zhang**, Z. Wu, and W. Hua, "Design of *n*-layer flat wire winding in permanent magnet machine for electric vehicles," *IEEE Trans. Transport. Electrific.*, vol. 11, no. 3, pp. 7798–7810, 2025.

[J6] **W. Zhang**, Z. Wu, L. Jin, Y. Fan, W. Hua, and M. Cheng, "Analysis and multi-objective optimization of the hybrid excitation switched flux machine," *IEEE Trans. Ind. Appl.*, vol. 61, no. 2, pp. 2996–3006, Mar.–Apr. 2025.

[J7] **W. Zhang**, Z. Wu, Z. Wang, W. Hua, Y. Fan, Z. Xu, *et al.*, "Design tradeoff between flux regulation capability and DC winding induced voltage in hybrid excitation switched flux machine," *IEEE Trans. Ind. Appl.*, vol. 61, no. 5, pp. 7198–7208, Sep. 2025.

[J9] **W. Zhang**, Z. Wu, Z. Cang, W. Hua, Y. Fan, and M. Cheng, "Influence of end-effect in switched flux machines," *IEEE Trans. Ind. Appl.*, vol. 62, no. 1, pp. 785–798, Jan. 2026.

[J10] **W. Zhang**, Z. Wu, and W. Hua, "High-fidelity lumped parameter thermal model for oil-cooling hairpin winding PM machines in EVs," *IEEE Trans. Transport. Electrific.*, vol. 12, no. 2, pp. 2978–2991, Apr. 2026.

[J11] **W. Zhang**, Z. Wu, and W. Hua, "Design of notching and mechanical stress relieving grooves for high-speed PM machine in EVs," *IEEE Trans. Ind. Appl.*, early access, 2026.

[J12] Z. Wu, **W. Zhang**, Y. Fan, W. Hua, and M. Cheng, "A transient-state lumped parameter thermal model for brushless wound field switched flux machines," *IEEE Trans. Transport. Electrific.*, vol. 10, no. 1, pp. 224–238, Mar. 2024.

[J21] J. Chen, W. Hua, L. Shao, Z. Wu, X. Yan, and **W. Zhang**, "Modified magnetic equivalent circuit of double-stator single-rotor axial flux permanent magnet machine considering stator radial-end flux-leakage," *IET Electr. Power Appl.*, vol. 18, no. 2, pp. 195–207, Feb. 2024.

[J22] Z. Wu, X. Yan, Z. Wang, **W. Zhang**, and W. Hua, "Influence of PWM excitation on loss and temperature of flat wire permanent magnet machines for electric vehicles," *IEEE Trans. Ind. Electron.*, vol. 73, no. 4, pp. 5174–5185, Apr. 2026.

[J23] Z. Wu, E. Fu, J. Du, **W. Zhang**, and W. Hua, "Quantitative comparison of electromagnetic vibration of PMSMs with glued and welded stator core for electric vehicles," *IEEE Trans. Ind. Appl.*, early access, 2025.

[J25] Z. Cang, **W. Zhang**, Z. Wu, and W. Hua, "Open-circuit magnetic equivalent circuit model for single-side double-row stator axial flux wound field machine," *IET Electr. Power Appl.*, vol. 20, no. 1, Art. no. e70216, 2026.

[C1] **W. Zhang**, Z. Wu, and W. Hua, "A deep learning based magnetic field surrogate modeling method for interior permanent magnet machine in EVs," in *Proc. Int. Conf. Elect. Mach. (ICEM)*, Funchal, Madeira, Portugal, 2026, accepted.

[C2] **W. Zhang**, Z. Wu, and W. Hua, "Comparative investigation of bridge-retained and carbon fiber sleeved interior permanent magnet machines in electric vehicles," in *Proc. Int. Conf. Elect. Mach. Syst. (ICEMS)*, Singapore, 2026, accepted.

[P2] Z. Wu, **W. Zhang**, L. Jin, and W. Hua, "Disk brushless electric excitation magnetic field modulation motor with double-row stator structure," CN118100572A, 2024.

[P3] Z. Wu, **W. Zhang**, H. Huang, and W. Hua, "Method for constructing a high-precision motor performance database based on a dual-model strategy," CN117972924A, 2024.

[P4] Z. Wu, **W. Zhang**, W. Hua, and Z. Zou, "Method for establishing general lumped parameter thermal network model of motor," CN117094088A, 2023.